

# KPP Validation Report

## MITgcm vs Python Port Comparison

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MITgcm Reference:

File: mitgcm\_kpp\_outputs\_seaice\_obcs\_1dmix034.nc  
UUID: dd828b05-e054-4089-848a-829a7f1ad441

Python Port Output:

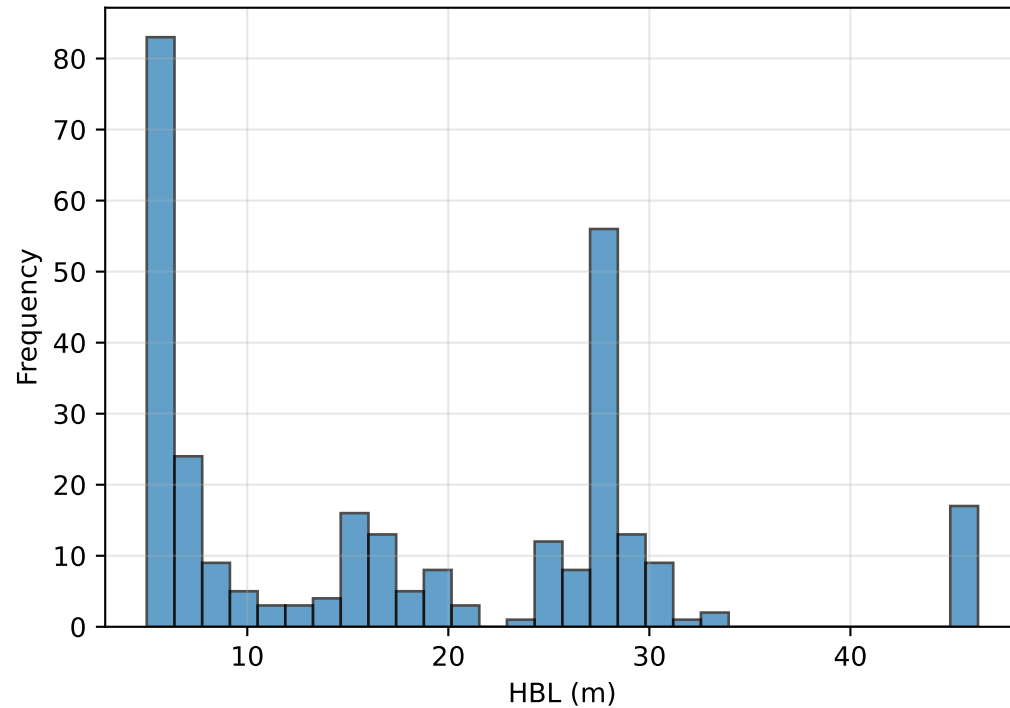
File: python\_kpp\_outputs\_seaice\_obcs\_1dmix034.nc  
UUID: dd828b05-e054-4089-848a-829a7f1ad441

Provenance: ☐ UUIDs match

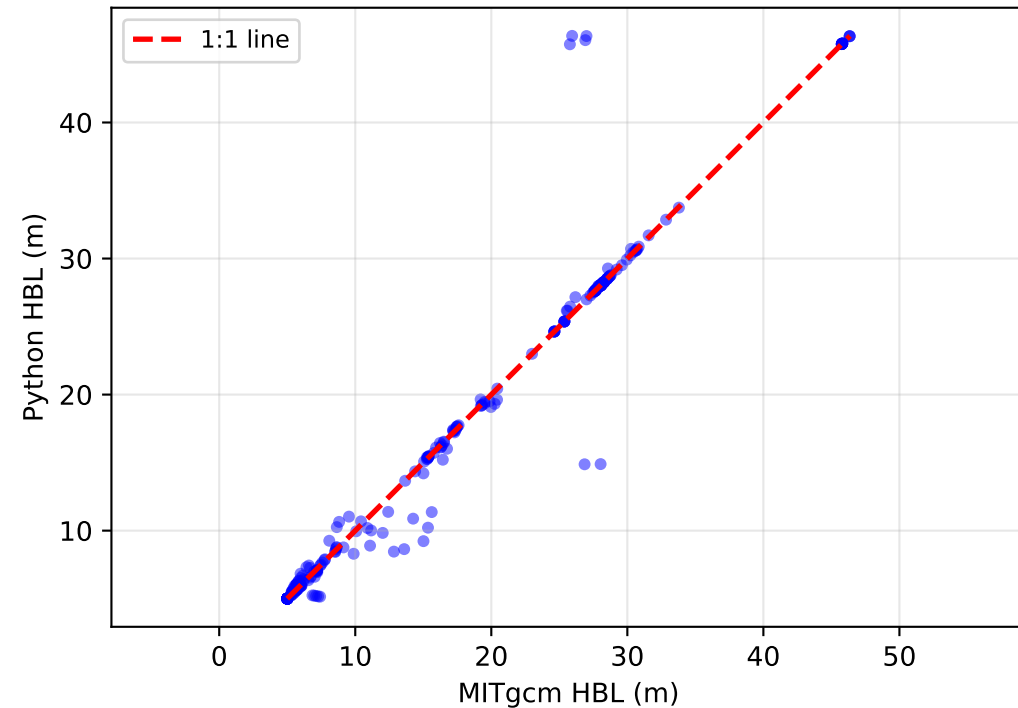
Timesteps: 5  
Grid: 10 × 8 × 23

# Boundary Layer Depth (HBL) Comparison

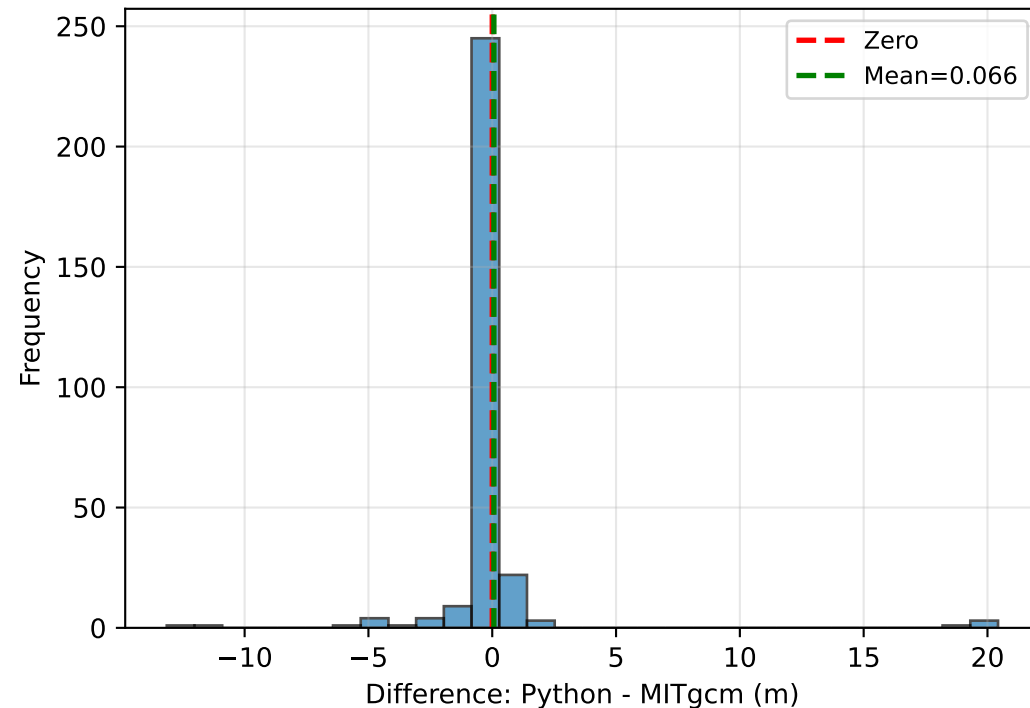
## MITgcm HBL Distribution



## HBL Scatter Plot



## HBL Difference Distribution



### HBL Statistics:

Mean |diff|: 0.627738 m  
Max |diff|: 20.425466 m  
RMS diff: 2.645172 m

Mean rel error: 3.9299%  
Median rel error: 0.1541%

Points compared: 295

### Range:

MITgcm: 5.0 – 46.3 m  
Python: 5.0 – 46.4 m

### Assessment:

△ □ CHECK

# Mixing Coefficients Statistics

Active Mixing Regions (coefficient > 1e-6 m<sup>2</sup>/s)

=====

VISC\_AZ: □ EXCELLENT

|                        |                                |
|------------------------|--------------------------------|
| Median relative error: | 0.000000%                      |
| Mean relative error:   | 287.465233%                    |
| Max relative error:    | 39863.932387%                  |
| 95th percentile error: | 60.419780%                     |
| Median absolute diff:  | 0.000000e+00 m <sup>2</sup> /s |
| Points compared:       | 3510                           |

DIFF\_KZ\_S: □ EXCELLENT

|                        |                                |
|------------------------|--------------------------------|
| Median relative error: | 0.000000%                      |
| Mean relative error:   | 401.461919%                    |
| Max relative error:    | 89787.285068%                  |
| 95th percentile error: | 70.815724%                     |
| Median absolute diff:  | 0.000000e+00 m <sup>2</sup> /s |
| Points compared:       | 3510                           |

DIFF\_KZ\_T: □ EXCELLENT

|                        |                                |
|------------------------|--------------------------------|
| Median relative error: | 0.000000%                      |
| Mean relative error:   | 401.461919%                    |
| Max relative error:    | 89787.285068%                  |
| 95th percentile error: | 70.815724%                     |
| Median absolute diff:  | 0.000000e+00 m <sup>2</sup> /s |
| Points compared:       | 3510                           |

GHAT: □ GOOD

|                        |             |
|------------------------|-------------|
| Median relative error: | 0.571382%   |
| Mean relative error:   | 9.173732%   |
| Max relative error:    | 155.291283% |
| Points compared:       | 209         |

# Validation Summary & Assessment

## OVERALL VALIDATION RESULT:

Status: ☒ NEEDS REVIEW

## COMPONENT RESULTS:

|                             |                                           |
|-----------------------------|-------------------------------------------|
| HBL (Boundary Layer Depth): | <input checked="" type="checkbox"/> CHECK |
| Max difference:             | 20.425466 m                               |
| Mean relative error:        | 3.9299%                                   |
| VISC_AZ:                    | <input type="checkbox"/> EXCELLENT        |
| Median relative error:      | 0.000000%                                 |
| DIFF_KZ_S:                  | <input type="checkbox"/> EXCELLENT        |
| Median relative error:      | 0.000000%                                 |
| DIFF_KZ_T:                  | <input type="checkbox"/> EXCELLENT        |
| Median relative error:      | 0.000000%                                 |
| GHAT:                       | <input type="checkbox"/> GOOD             |
| Median relative error:      | 0.571382%                                 |

## PROVENANCE:

|              |                                     |
|--------------|-------------------------------------|
| UUID Match:  | <input type="checkbox"/> Yes        |
| MITgcm UUID: | dd828b05-e054-4089-848a-829a7f1a... |
| Python UUID: | dd828b05-e054-4089-848a-829a7f1a... |

## ACCEPTANCE CRITERIA:

|                                            |                                  |
|--------------------------------------------|----------------------------------|
| <input type="checkbox"/> EXCELLENT:        | HBL < 0.1m, mixing < 0.1% error  |
| <input type="checkbox"/> GOOD:             | HBL < 1.0m, mixing < 1.0% error  |
| <input checked="" type="checkbox"/> CHECK: | Exceeds thresholds - investigate |